Thick Cloud Removal With Optical and SAR Imagery via Convolutional-Mapping-Deconvolutional Network

Wenbo Li, Ying Li, and Jonathan Cheung-Wai Chan

***Abstract*— In this article, we proposed a thick cloud removal method for remote-sensing imagery based on multisource estimation. A convolutional-mapping-deconvolutional (CMD) network is proposed to estimate the cloud-free image directly from multisource reference images. Synthetic aperture radar (SAR) image and low-resolution heterogeneous (LRH) image, namely, image from a different optical sensor with lower spatial resolution, are used as reference images to recover the missing information in the cloud-contaminated high-resolution (HR) image. The CMD net is composed of three functional components: the convolutional layers for encoding, the mapping layer for feature transferring, and the deconvolutional layers for decoding. In the training procedure, HR images from cloud-free regions and their corresponding LRH and SAR reference images are used to train the CMD net. When the CMD net is fully trained, it is able to estimate the HR images with their corresponding LRH and SAR reference images. The LRH and SAR reference images are first encoded by the convolutional layers before being transferred to the feature space at HR by the mapping layer. The transferred features are then decoded into cloud-free HR image by the deconvolutional layers. Cloud-free regions in the cloud-contaminated HR image are used to further improve the estimated image via intensity normalization. At last, the cloudy pixels are replaced by their corresponding pixels from the estimated cloud-free HR image. Comparisons with several recently proposed multisource cloud removal methods show that our proposed method is superior as validated by quantitative indexes and visual inspections.**

***Index Terms*— Cloud removal, convolutional neural network (CNN), intensity normalization, multisource estimation.**

I. Introduction

CLOUD cover has severed the useful information in optical remote-sensing imagery. According to statistics, about one-third of the Earth's observation images are contaminated by cloud [1]. When cloud percentage coverage is high, surface information goes missing, causing significant data gaps. Thick clouds blatantly blocked all the radiation signals. Therefore, the removal of thick cloud from optical remote sensing imagery is considered one of the most important issues in optical remote sensing. Existing methods for thick cloud removal can be divided into three categories: spatial-based, multitemporal-based, and multisource-based [2].

Spatial-based methods come from the inpainting methods for natural images restoration. These methods estimate the pixels in cloud regions with information obtained from cloud-free regions in the same remote sensing image. Many different techniques have been utilized to exploit the spatial correlation in the cloud-contaminated image. For example, in [3], bandelet transform and multiscale grouping are used to represent the multiscale geometry of image structure. In [4], the missing information is reconstructed by propagating the spectral–geometrical information retrieved from the cloud-free parts. Meng *et al.* [5] recover the missing information patch by patch via sparse representation using feature dictionary learned with exemplars from the cloud-free regions. When the cloud regions are small and scattered, spatial-based method is able to reconstruct the missing information with no additional reference image. However, the performance of these methods is poor when cloud regions are large.

Multitemporal-based methods take advantage of the revisiting property of orbital acquisitions. The missing information in a cloudy region is estimated with cloud-free correspondence in reference images obtained at other times by the same sensor. In [1] and [6], information cloning is used to eliminate the difference brought by seasonal changes between the cloud-contaminated image and the temporal reference image. In [7], the information from a temporal reference image is used as guidance to find similar pixels in the cloud-free regions to replace the cloudy pixels. Robust principle component analysis (RPCA) framework is adopted in [8] and [9] to handle the observed cloud image sequence of the same area. Method proposed in [10] expanded the utilization of iterative haze-optimized transformation (IHOT), making it possible to remove thick and thin clouds at the same time. The contamination of thick clouds is regarded as spatial gaps in [6] and gets removed via harmonic analysis of time series with true value constraint. A sparse representation-based approach is proposed by Li *et al.* [11] to recover the missing information with a multitemporal learned dictionary. Besides surface reflectance, multitemporal reference can also be utilized to recover other

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information corrupted by thick clouds; for example, it is used to restore land surface temperature (LST) in [12]. Multitemporal methods usually perform well when the time gap between the target and the reference images is short and no significant change occurs during this time gap. However, for some high-resolution (HR) optical satellite missions, short revisiting period is not guaranteed, limiting the future development of multitemporal methods.

Multisource-based methods are recently proposed to take advantage of data obtained by other remote sensors to estimate the missing information in optical remote sensing imagery. In some approaches [13]–[15], images obtained by other optical sensors with higher temporal resolution are used as auxiliary to shorten the time gap between cloud-contaminated image and cloud-free reference. However, it should be noted that there is tradeoff between spatial resolution and temporal resolution [14]; a higher temporal resolution source usually has lower spatial resolution [13], which will hamper the recovery of missing information. On the other hand, although using other optical sources can make the time gap shorter, certain time gap still exists between cloudy image and optical reference, because the reference source is also optical and will be influenced by thick cloud contamination. Some other researchers utilize synthetic aperture radar (SAR) sources [16]–[18] as reference, the strong penetration ability of active microwave free SAR from cloud contamination, making it possible to get cloud-free reference on the same day of the original image, and in most cases, the spatial resolution of SAR sensors is also high enough. However, the difference in imaging mechanism between active and passive sensors still limits the adoption of method that utilizes SAR data sources. Methods proposed in [19] and [20] use optical image and SAR image as reference simultaneously and recover the information in cloud regions via sparse representation. Experimental results in [19] and [20] show that using SAR and optical reference together outperforms using either of them separately.

In general, for thick cloud removal, multisource reference data show two major advantages.

First, the acquisition time gap between the cloud-contaminated image and the multisource reference image can be much shorter compared to multitemporal reference. The time gap between cloud-contaminated image and multitemporal reference image is at least one revisiting period, and in some unfortunate cases, when images are not at good quality, it would take even longer. Longer time gap means more chance of changes, making the reference data less reliable. Since multisource reference data are acquired by different sensors, revisiting period is less as an obstacle. The possible time gap between a cloud-contaminated image and multisource reference images can be several days, sometimes even just several hours.

Second, reference images acquired by active sensors such as SAR images are free from cloud contamination. Although thick clouds block radiation from the Earth surface, they can be penetrated by the microwaves from active sensors. The SAR reflectance from cloudy regions provides us real-time information even under thick cloud cover. The reliability of this information is higher than any other references.

On the other hand, there are also two big challenges in utilizing multisource reference data. Different imaging mechanisms may result in different representations of the same scene. Optical imagery represents the Earth surface with the intensity of radiation, while the SAR representation is determined by the intensity of reflectance of microwaves. Although the structural components in the optical and SAR images from the same scene are very similar, for example, lines for road and homogeneous regions for water, significant differences can still be witnessed in pixel intensity. In addition, optical sensors may also differ from each other due to different spatial and spectral configurations. To harmonize scales and exploit different configuration of sensors are major challenges in using multisource reference data for the reconstruction of missing information.

Latest novel methodologies such as sparse representation [11], [19], [20] and deep learning [21], [22] have already been applied to cloud removal. These state-of-the-art methods are very effective in recovering the missing information in the cloud-contaminated regions. In this article, we will compare our method with those latest methods based on sparse representation and deep learning.

The main contributions of this work are summarized as follows.

1) In the proposed method, the cloud-free HR image is estimated directly from multisource reference images via a convolutional neural network (CNN) named convolutional-mapping-deconvolutional (CMD) net. The CMD net conducts an encoding-mapping-decoding operation from low-resolution heterogeneous (LRH) and SAR reference images to cloud-free HR image with three functional components: the convolutional layers, the mapping layer, and the deconvolutional layers. To the authors' knowledge, although some related works have been done to make use of spatial–temporal–spectral information to recover missing information, this is the first application of CNN in multisource-based cloud removal for optical imagery.
2) The CMD net expanded the application of CNN from image restoration to multisource estimation. The input and output images are acquired by different remote sensors, so that they are actually represented in different feature spaces. Both the reference data (Landsat-8 and Sentinel-1) and the target data (Sentinel-2) are from real remote sensing sources. This work does not only provide one solution for thick cloud removal but also show that the multisource information can be more helpful in recovering the missing information in remote sensing images.
3) We tested the proposed method on both simulated and real cloud contaminations and made comparisons with several other recent multisource methods. The result demonstrated that the proposed method outperforms other multisource methods significantly in both quantitative indexes and visual inspections. Testing samples from another scene with land cover types not included in the training samples are used to validate the robustness of CMD net. The experiments on parameters

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also demonstrated that smaller convolutional kernels and higher learning rate will benefit the performance of estimation. By testing more matching bands between Landsat-8 and Sentinel-2 sensors, we demonstrated that CMD net is able to handle more bands without degrading the performance of original bands.

The rest of the article is organized as follows. Section II gives an introduction to some related works. The proposed method is discussed in Section III. Section IV discusses the experimental results. Section V details the discussion on the advantages and disadvantages of proposed method. At last, conclusions are drawn in Section VI.

II. Related Works

A. Multisource Thick Cloud Removal

Many different Earth observation missions have been launched to monitor the Earth surface. For most locations on Earth, it is possible to get multiple corresponding remote sensing images acquired by different sensors within a short period of time. Although these sensors may have different imaging mechanisms and characteristics, they are different representations of the same ground surface, and there is inherent relationship in structural components between them. For example, the same road may have different intensity in multisource remote sensing images, but the same linear structures can still be witnessed.

Depending on their ways of utilizing multisource references, multisource cloud removal methods can be divided into two major categories: exemplar-based and estimation-based. The exemplar-based methods take reference images as structural guidance and interpolate the cloudy regions with pixels [16] or patches [18] from cloud-free regions in the cloud-contaminated image. These methods utilized the property of nonlocal similarity and structural redundancy in remote sensing imagery. Since the information of cloudy pixels is totally missing, it would be very difficult to find their analogs in the cloud-free regions. Therefore, the structural information in multisource references is used for similarity matching. For each patch in the cloudy region, its similar patches are selected based on the similarity of their corresponding patches in the reference image, and then, the similar patches are used to replace the corresponding cloudy patches. These methods mainly focus on the sequence of similar patch searching and priority of patch replacing. For example, in [16], searching for similar patches is designed to follow a spiral order from center to the neighborhood of cloudy patch. Nearby similar patches would be selected with higher priority instead of those far away. The central pixel of most similar patch is used to replace the central pixel of cloudy patch. In [18], the priority of each cloudy patch to be replaced is decided by both its distance to cloud-free regions and its complexity of structure information. Patches on the boundaries of different landscapes are replaced first, and then, the pixels in them are used as reference to search similar patches of nearby patches containing relatively monotonous landscapes. On the one hand, exemplar-based methods are relatively simpler to use, and the restored information is similar to the cloud-free regions statistically in intensity. On the other hand, exemplar-based methods usually fail to restore the missing information accurately, especially when there are little similar structural components in the cloud-free regions.

The estimation-based methods take reference image as input to estimate the missing information in the cloudy regions. For example, the method proposed in [19] and [20] estimates the missing information in cloudy regions via sparse representation. In these methods, structural correspondence is established between reference images and target image. Instead of seeking for analogs from the cloud-free regions in the original cloud-contaminated image, the missing information is estimated from their corresponding reference data. These methods usually focus on establishing structural correspondence of data from different sources. For example, in [19], the missing information to be restored was decomposed into two different parts: the high-frequency part and the low-frequency part. The high-frequency part corresponds to detailed features such as lines and points, whereas the low-frequency part corresponds to contours and homogeneous regions. Dictionary pair learning (DPL) is used to build structural correspondence between SAR reference data and the high-frequency part of missing information, while correspondence for the low-frequency part is estimated with low-resolution reference data. The high-frequency part and the low-frequency part are restored separately with sparse representation, and then, they are combined into cloud-free information. Although in [20] the missing information is no longer decomposed, dictionary group learning (DGL) is utilized to establish structural correspondence between the missing information and two different types of reference data, namely, the SAR and the low-resolution reference data, under one single framework. The missing information is restored with sparse representation and the constraint of nonlocal similarity. Compared to exemplar-based methods, estimation-based methods are able to perform better when there are little structural components similar to the missing information in the cloud-free regions. However, the estimated information may not be as statistically similar to the cloud-free regions in intensity as the exemplar-based results.

B. CNN for Image Restoration

Inspired by the success of CNN in the field of natural image classification [23]–[25], some CNN-based methods have been proposed for remote sensing image classification [26]–[28] as well as restoration of natural images [29], [30] and remote sensing images [21], [22]. An architecture named fully convolutional network (FCN) [31] is often adopted to solve restoration problems. In the FCN architecture, fully connected layers are removed to ensure a 2-D output for each 2-D input. While FCN is applied to image restoration, pooling layers are usually removed to avoid the loss of information too.

In order to better restore the missing information in natural images, many different techniques are introduced into CNN-based image restoration for different restoration purposes. For example, RED-net [29] introduced

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"skip connections" into image inpainting to avoid information loss in the encoding–decoding procedure. In image inpainting, information is assumed completely missing in the damaged regions. However, the undamaged regions do not suffer from any information loss. On the one hand, an encoding–decoding FCN is able to reconstruct the missing information in the damaged regions with the reference of neighbor regions. On the other hand, the undamaged regions would also suffer from some information loss during the encoding–decoding procedure. The "skip connections" transfer information from the convolutional layers to their symmetric deconvolutional layers, so that the original information in the undamaged regions is preserved and information loss is avoided.

In [30], residual blocks are built to extract the additive noises in the image denoising task. In image denoising, the whole image is usually assumed to be contaminated by additive white Gaussian noise. These noises correspond to high-frequency corruptions in the frequency domain so that the original information corresponding to high-frequency suffers most from them, while the original low-frequency information suffers little. In each residual block, connections are established between the input and output, so that the convolutional layers within the block can focus on restoring the high-frequency information interrupted by noise, while low-frequency information can be directly provided by the input. In this way, the utilization efficiency of kernels in convolutional layers is improved significantly, leading to better restoration of information.

One of the most interesting tricks is the mapping layer adopted in SRCNN [32]. Instead of simply modeling the task of image super-resolution as the restoration of high-frequency components, SRCNN uses super-resolution as a mapping from the low-resolution feature space to HR feature space. A mapping layer with 1×1 convolutional kernels is used to conduct the feature transferring. Although using convolutional layers for both encoding and decoding limited the potential of further deepening of SRCNN, the idea of mapping layer and feature transferring can be extended further to the estimation between different remote sensing sources.

III. Proposed Method

In our proposed method, we further extended CNN image restoration to multisource estimation. Fig. 1 shows the procedures of the proposed method. First, the CMD net is trained with training data from cloud-free regions in the original remote sensing image. One set of training samples includes an LRH image and an SAR image as input, and a cloud-free HR image as target image. A fully trained CMD net is expected to take the LRH and SAR image as input and produce the corresponding cloud-free HR image from the same training set as output. When the CMD net is fully trained, it can be used to do cloud removal on the testing data. One set of testing samples includes an LRH image and an SAR image as input, and a cloud-contaminated HR image. To estimate the corresponding cloud-free HR image of the cloud-contaminated HR image in a set of testing sample, the LRH image and the SAR image from the same set of testing sample are used as input for the fully trained CMD net. Then, the cloud-free regions in the cloud-contaminated HR image are utilized to improve the estimated result via intensity normalization. At last, the cloudy pixels in the cloud-contaminated HR image are replaced by their corresponding pixels in the normalized image. In this section, we will discuss the details about our proposed method, including network architecture, network training, and intensity normalization.

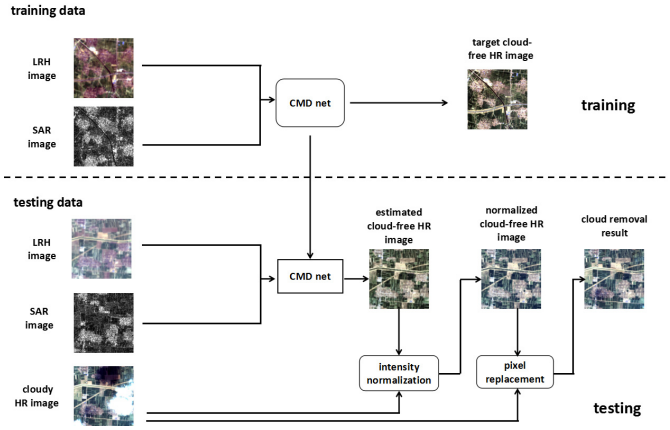


Fig. 1. Flowchart of procedures in our proposed method.

A. Architecture of CMD Network

In this section, we will discuss the architecture of CMD net. We mainly focused on the design and verification of three functional components in the CMD net: the convolutional layers for encoding, the mapping layer for feature transferring, and the deconvolutional layers for decoding.

In the proposed method, a CMD net is used to accomplish the multisource estimation from reference images to target the cloud-free image. Different from natural image restoration, the input images of the CMD net is no longer a defective version of output images but totally heterogeneous representations of the same scene. As we have mentioned in Section I, the multisource input images may differ from the target output image in spatial resolution, bandwidth, and range, and even representation patterns. In order to solve the problem of multisource estimation, we designed the CMD net as a sequential combination of three functional components: the convolutional layers, the mapping layer, and the deconvolutional layers. In this framework, multisource input images are first encoded by the convolutional layers; then, the mapping layer transfers the encoded features to the feature space of output image; at last, the transferred features are decoded into cloud-free output image by deconvolutional layers.

The architecture of CMD net is as shown in Fig. 2. The network we used for experiments is composed of nine layers: four convolutional layers, one mapping layer, and four deconvolutional layers. The amount of feature maps in each middle layer (conv1 to deconv3) is 64. The layer of deconv4 gives the estimated output. Kernels of 3×3 are used in convolutional and deconvolutional layers, while the kernel size in the mapping

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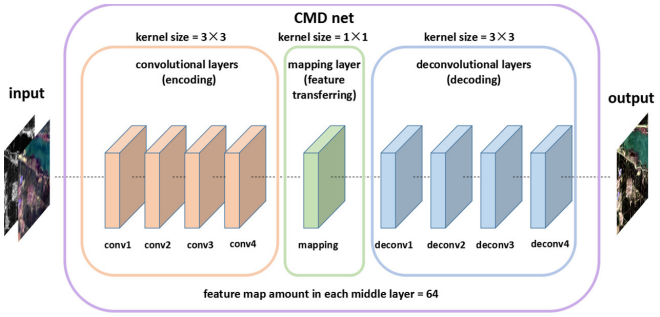


Fig. 2. Architecture of CMD net.

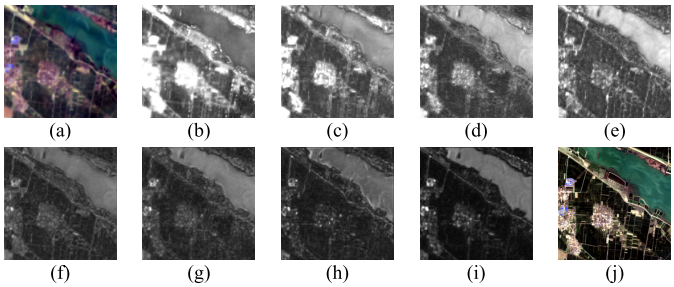


Fig. 3. Visualized demonstration of MRM in each middle layer. (a) LRH input image. (b) MRM of conv1 layer. (c) MRM of conv2 layer. (d) MRM of conv3 layer. (e) MRM of conv4 layer. (f) MRM of mapping layer. (g) MRM of deconv1 layer. (h) MRM of deconv2 layer. (i) MRM of deconv3 layer. (j) Result in deconv4 layer.

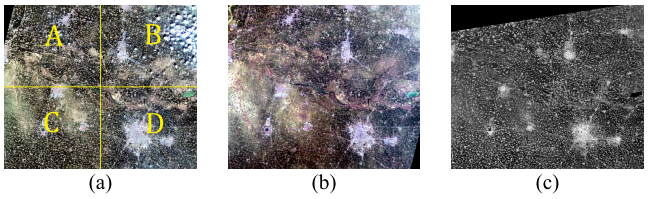


Fig. 4. Source images used for experiment. (a) HR source image. (b) LRH source image. (c) SAR source image.

layer is 1×1. The activation function between layers is rectified linear unit (ReLU) [23].

Since four layers with 3×3 kernels, two layers with 5×5 kernels, and one layer with 9×9 kernels share the same size of reception field in convolution and deconvolution, we used four convolutional layers and four deconvolutional layers with 3×3 kernels for experiment, so that comparisons can be easily made with the cases when kernel size is 5×5 or 9×9 under the same size of reception field. The amounts of convolutional and deconvolutional layers can be changed to fit the data scale in practice as long as they have the same number of layers.

Convolutional layer [33], deconvolutional layer [34], and ReLU activation function [23] are all classic elements in CNN, and their implementation details are sufficiently discussed in previous publications. Mapping layer [32] is a special type of convolutional layer, whose kernel size is 1×1. Almost all the deep learning tools (Caffe, Tensorflow, Pytorch, etc.) take them as basic functions. For their implementation details, we refer the reader to related documentations [23], [33], [34]. Instead, we would discuss their functions in the CMD net.

Considering that the feature maps in a fully trained CNN network are usually sparse and the feature extraction for each pixel is done by a few kernels. We select the maximum response for each pixel to represent its output value in all the feature maps from each middle layer. In this way, we are able to create a maximum response map (MRM) for each middle layer. The reason why we choose the maximum response instead of average response for each pixel is that the kernels in each layer are redundant and the high responses to similar kernels should be counted only once.

Fig. 3 gives a visual demonstration of MRM in each middle layer. It should be noted that the feature maps in each layer is the output data after convolution, namely, the mapping operation is conducted between Fig. 3(e) and (f), and the first deconvolution operation is conducted between Fig. 3(f) and (g). As shown in Fig. 3(b)–(e), an encoding procedure is done by the convolutional layers: the value of each pixel is determined by its response to certain structural components instead of its original digital number value. The pixels for roads are significantly enhanced since the line features in the input images are extracted by convolution kernels. Similarly, the pixels for water give high response to their corresponding kernels, although their digital value in the input images is small. Significant change can also be witnessed from Fig. 3(e) and (f): the intensity of almost all the pixels is eliminated after going through the mapping layer, which means the features have been transferred to the feature space of target output so that they can be represented with lower cost. At last, Fig. 3(f)–(j) demonstrates the decoding procedure: the transferred features are turned into output image by the deconvolutional layers. It should be noted that the feature map in Fig. 3(i) is very sparse, which means the kernels in the last deconvolutional layer have extracted the features successfully and became very representative.

It is verified that all three functional components have successfully accomplished their predefined tasks: the convolutional layers encoded the input images into features; then, the features are transferred to the output feature space by the mapping layer; at last, the transferred features are decoded into output image by the deconvolutional layers.

B. Training for CMD Network

In this section, we will discuss the training for the CMD net, including the selection of training samples, the input and output formulation of network, and the configuration of network training. Remote sensing images from three real sources are selected for network training and testing in this work. Landsat-8 is used as the LRH source, Sentinel-1 is used as the SAR source, and Sentinel-2 is used as the HR source. Details about the three sources used in this work are shown in Table I, including the acquisition date, spatial resolution, and the band centers for the three optical bands selected for experiments. The source images are demonstrated in Fig. 4, and their spatial dimensions are all $10\,980 \times 9362$. This scene is located near the Shangqiu City in China, in which the major land cover types are farmland and residential areas.

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TABLE I
DETAILS OF SOURCE DATA

Data Type	Data source	Acquisition Date	Spatial resolution	Band centers		
HR	Sentinel-2	2017.3.26	10m	R(B4)	G(B3)	B(B2)
				0.665 μm	0.560 μm	0.490 μm
LRH	Landsat-8	2017.2.16	30m	R(B4)	G(B3)	B(B2)
				0.655 μm	0.563 μm	0.483 μm
SAR	Sentinel-1	2017.3.26	10m			

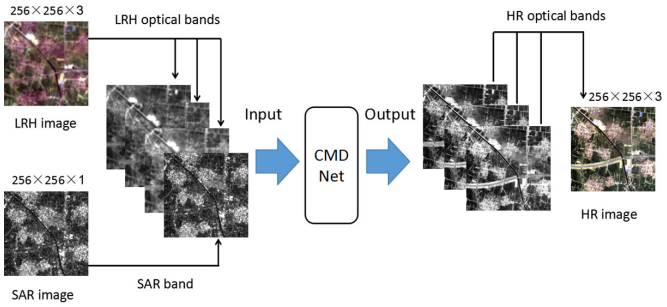


Fig. 5. Input and output for CMD net.

In order to better test the performance of network, the training and testing samples should be selected from different regions in the source images, i.e., HR image which is contaminated. As shown in Fig. 4, the HR image is divided into four regions (A–D). Region A contains a part with no data in both HR and SAR source images. Region B contains a part with no data in the SAR source image and is contaminated by thick clouds in HR source image. Region C is cloud-free in the HR source image. Region D contains a part with no data in LRH source images. Therefore, we select training samples from region C, and testing samples are selected from region B for the experiments of removing both simulated and real clouds. In this work, 1000 sets of training samples are selected from region C to train the network (data set downloadable: https://datahub.io/lwb19910620/dataset_tgrs-2018-01535-dull-seahorse-98).

The sampling of training data is made by window sliding with overlapping in spatial size 256×256. Then, 1000 cloud-free samples from region C are selected randomly as training samples. Each training sample includes an LRH image with three bands, an SAR image with one band, and an HR image with three bands. In the training procedure, the LRH and SAR images are taken as input of the CMD net, and the HR images are used as target; the difference between actual output of the CMD net and HR images is backpropagated through each layer to adjust the parameters in the CMD net to make the output of CMD net as close to HR images as possible.

In this work, we established an end-to-end mapping from input to output. As shown in Fig. 5, three LRH optical bands and one SAR band are combined into a 3-D block as input, while the output is an equivalent 3-D block composed of three HR optical bands. The size of each band in both input and output is 256×256. In this work, only three visible bands are used for network training, since their corresponding bandwidth are similar in Landsat-8 and Sentinel-2 data.

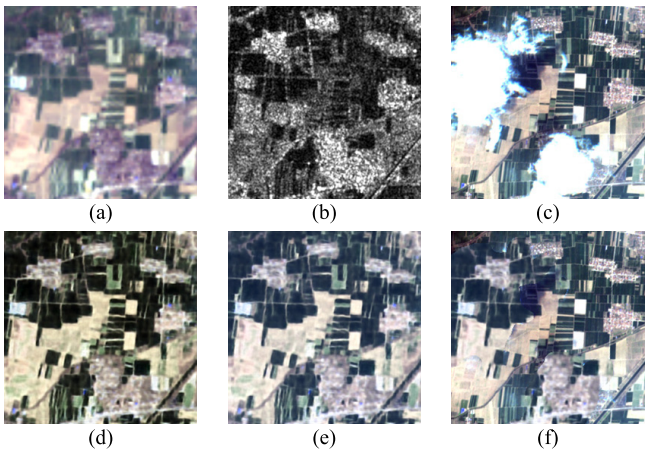


Fig. 6. Visual demonstration of intensity normalization and pixel replacement. (a) LRH reference image. (b) SAR reference image. (c) Cloud-contaminated HR image. (d) Estimated cloud-free image. (e) Normalized cloud-free image. (f) Cloud removal result after pixel replacement.

The procedure of network training is to learn the parameters Θ in network, including the weight of convolutional and deconvolutional kernels and the bias of each layer. This is done by minimizing the Euclidean distance between the output of network and target image. Given Y_i as the input composed of LRH and SAR bands, and X_i as the expected cloud-free HR target image, the following mean squared error (MSE) is minimized:

$$L(\Theta) = \frac{1}{N}\sum_{i=1}^{N} \|F(Y_i; \Theta) - X_i\|^2. \tag{1}$$

The implementation of the training procedure is made with Tensorflow, and stochastic gradient descent (SGD) is adopted to minimize the loss function. The learning rate is set to 0.05.

C. Intensity Normalization

In most cases, the cloudy image is only partially contaminated by thick clouds, which means that the cloud-free pixels can also be used as reference to further improve the estimated results. In this work, intensity normalization [35] is adopted to take advantage of cloud-free pixels for further improvement of the expected cloud-free HR images.

As shown in formulation (2), $I_E(i, j)$ is the pixel on the (i, j) position in the estimated image I_E, while $I_N(i, j)$ is its corresponding pixel in normalized image I_N. P_f is the set of cloud-free pixels in cloud-contaminated HR image, and $\mu(P_f)$ and $\sigma(P_f)$ are its mean and variance, respectively. P_e is the set of pixels corresponding to P_f in I_E, and $\mu(P_e)$ and $\sigma(P_e)$ are its mean and variance, respectively,

$$I_N(i, j) = (I_E(i, j) - \mu(P_e)) \times \frac{\sigma(P_f)}{\sigma(P_e)} + \mu(P_f). \tag{2}$$

Once the intensity normalization is done, the corresponding pixels in the normalized image are used to replace the cloudy pixels in the cloud-contaminated HR image. The results of intensity normalization and pixel replacement are shown in Fig. 6: (a) is the LRH reference image; (b) is the SAR

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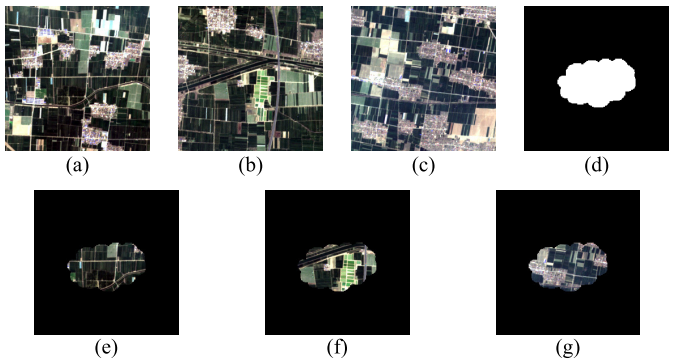


Fig. 7. Cloud-free HR images and cloud mask for simulated cloud removal. (a) Cloud-free HR image A. (b) Cloud-free HR image B. (c) Cloud-free HR image C. (d) Cloud mask. (e) Masked land covers in image A. (f) Masked land covers in image B. (g) Masked land covers in image C.

reference image; (c) is the cloud-contaminated HR image; (d) is the estimated cloud-free HR image produced by the CMD net taking (a) and (b) as input; (e) is the normalized result of (d) using pixels from cloud-free regions in (c); and (f) is the cloud removal result after replacing cloudy pixels in (c) with their corresponding pixels in (e). In the cloud-free regions, it can be verified that intensity normalization improved the similarity between estimated image and original image in intensity.

IV. Experimental Results

Experiments of both simulated and real cloud removal are carried out. The results are compared with four recently proposed cloud removal methods [18]–[21] in Sections IV-A and IV-B for removing simulated clouds and real clouds, respectively. The method proposed in [19] is indicated with DPL method. DGL method and SAR structure data (SARSD) method are used to indicate the methods proposed in [18] and [20], respectively. The spatial-temporal-spectral deep convolutional neural network (STS-CNN) is used to indicate the method proposed in [21]. It should be noted that although only SAR reference image is used for "data term" in SARSD method, we used both LRH reference image and SAR reference image for "data term" in our experiments. In order to demonstrate the contribution of LRH source and SAR source separately, we added the results with only the LRH image as input and the results with only SAR image; they are indicated with CMD-LRH and CMD-SAR, respectively. CMD-LRH&SAR is used to indicate the results with both LRH image and SAR image as input. In Section IV-C, the robustness of CMD net is tested with samples from a different scene with land cover types not included in the training samples. Experiments on the influence of important parameters are carried out, and the result is discussed in Section IV-D. The performance of CMD net on more matching bands between Landsat-8 and Sentinel-2 is tested in Section IV-E.

A. Experimental Results on Removing Simulated Cloud

The simulated cloud-contaminated images are obtained by masking the assumed cloud regions on a cloud-free HR image. Fig. 7 shows three HR images used for experiments and the mask of cloud. Fig. 7(a)–(c) are three cloud-free HR optical images selected from region B in Fig. 4(a). They are assumed to be contaminated by thick cloud as shown in Fig. 7(d), and all the information within the cloud mask is missing. Fig. 7(e)–(g) are the corresponding masked land covers in Fig. 7(a)–(c). It can be seen that besides the common farmland land cover, three different land cover types are also included in each masked region, respectively: road in Fig. 7(e), river in Fig. 7(f), and residential areas in Fig. 7(g).

Since the cloud-contamination is simulated and the ground truth is available, we are able to evaluate the performance of different methods with peak signal to noise ratio (PSNR) and structural similarity (SSIM) [36] indexes. Figs. 8–10 demonstrate the cloud removal results. The corresponding PSNR

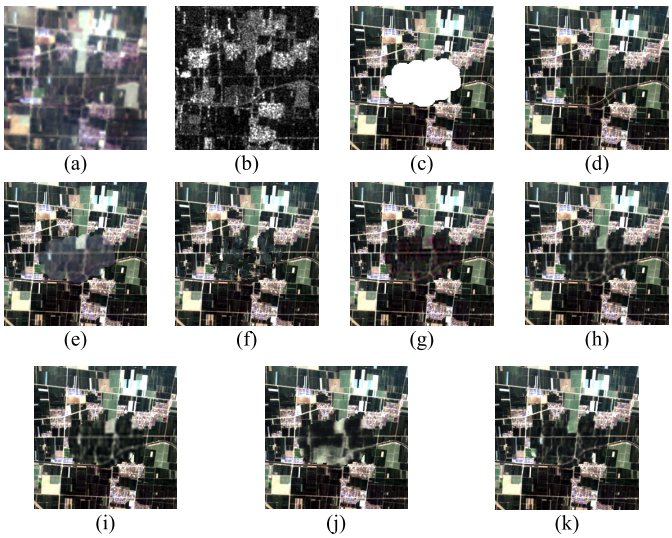


Fig. 8. Results of different methods in simulated cloud removal on image A. (a) LRH image. (b) SAR image. (c) Simulated cloudy HR image. (d) Ground truth. (e) Result of DPL. (f) Result of SARSD. (g) Result of DGL. (h) Result of STS-CNN. (i) Result of CMD-LRH. (j) Result of CMD-SAR. (k) Result of CMD-LRH&SAR.

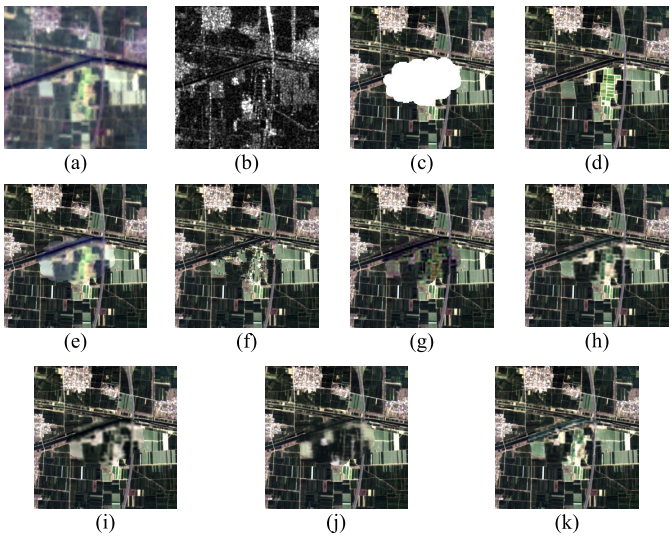


Fig. 9. Results of different methods in simulated cloud removal on image B. (a) LRH image. (b) SAR image. (c) Simulated cloudy HR image. (d) Ground truth. (e) Result of DPL. (f) Result of SARSD. (g) Result of DGL. (h) Result of STS-CNN. (i) Result of CMD-LRH. (j) Result of CMD-SAR. (k) Result of CMD-LRH&SAR.

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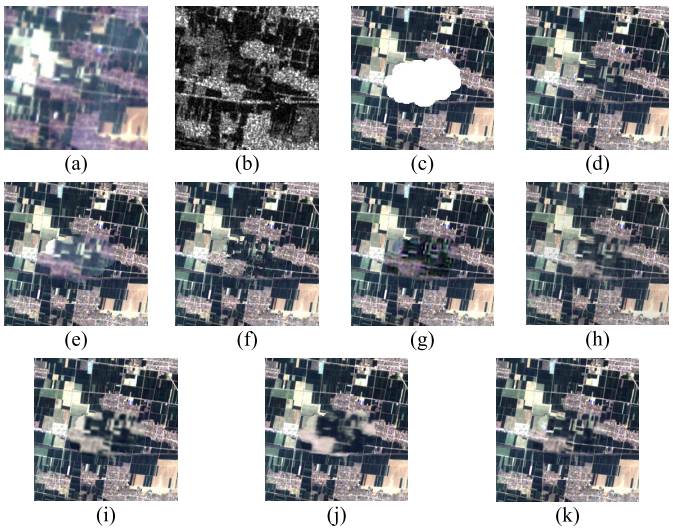


Fig. 10. Results of different methods in simulated cloud removal on image C. (a) LRH image. (b) SAR image. (c) Simulated cloudy HR image. (d) Ground truth. (e) Result of DPL. (f) Result of SARSD. (g) Result of DGL. (h) Result of STS-CNN. (i) Result of CMD-LRH. (j) Result of CMD-SAR. (k) Result of CMD-LRH&SAR.

TABLE II
QUANTITATIVE EVALUATION FOR THE RESULTS IN FIG. 8

		DPL	SARSD	DGL	STS -CNN	CMD -LRH	CMD -SAR	CMD-LRH &SAR
PSNR	B4	23.52	23.50	26.64	26.57	24.95	20.32	**27.30**
	B3	23.70	24.10	25.06	26.89	25.55	20.21	**27.75**
	B2	22.58	24.71	26.87	27.61	26.54	21.20	**28.49**
SSIM	B4	0.9022	0.8939	0.9268	0.9297	0.9085	0.8730	**0.9392**
	B3	0.9009	0.8922	0.9097	0.9289	0.9078	0.8730	**0.9369**
	B2	0.8969	0.8942	0.9184	0.9314	0.9127	0.8775	**0.9385**

TABLE III
QUANTITATIVE EVALUATION FOR THE RESULTS IN FIG. 9

		DPL	SARSD	DGL	STS -CNN	CMD -LRH	CMD -SAR	CMD-LRH &SAR
PSNR	B4	21.37	21.44	21.05	22.66	21.95	19.25	**23.39**
	B3	21.61	21.97	20.39	23.33	22.33	19.05	**23.83**
	B2	21.07	23.23	22.50	**24.57**	23.92	21.35	24.44
SSIM	B4	0.9007	0.8968	0.9118	0.9101	0.9017	0.8790	**0.9219**
	B3	0.9023	0.8987	0.9063	0.9128	0.9053	0.8814	**0.9219**
	B2	0.8919	0.8981	0.9090	0.9130	0.9069	0.8839	**0.9209**

TABLE IV
QUANTITATIVE EVALUATION FOR THE RESULTS IN FIG. 10

		DPL	SARSD	DGL	STS -CNN	CMD -LRH	CMD -SAR	CMD-LRH &SAR
PSNR	B4	23.16	24.18	23.97	24.95	24.25	22.81	**27.10**
	B3	23.08	24.86	24.16	25.61	24.79	23.12	**27.44**
	B2	23.41	26.21	25.56	26.44	26.41	24.74	**28.87**
SSIM	B4	0.9094	0.9082	0.9217	0.9257	0.9076	0.8983	**0.9390**
	B3	0.9066	0.9069	0.9182	0.9318	0.9063	0.8963	**0.9365**
	B2	0.9107	0.9087	0.9247	0.9351	0.9117	0.9028	**0.9389**

and SSIM values for Figs. 8–10 are listed in Tables II–IV, respectively. Higher value of PSNR and SSIM means better reconstruction performance. The highest values are in boldface.

It can be seen that the SARSD method scored high in PSNR, and the reconstructed missing information fits the cloud-free regions well. However, many detailed structures are not reconstructed correctly, resulting in poor performance measured by the SSIM index. On the contrary, the DGL method is able to recover the structural details due to its utilization of nonlocal similarity, while the reconstructed missing information shows to be less similar to the cloud-free regions. It should be noted that the recently proposed STS-CNN method has been shown to improve the result of cloud removal significantly; it scored the highest PSNR value in band 2 of the image shown in Fig. 9.

Compared to other multisource methods, the proposed method shows significant advantage in both visual effect and quantitative indexes. The reconstructed missing information shows good resemblance to the cloud-free regions, and most recovered details are rather accurate. Our method also shows significant advantage in the scores in quantitative indexes with the highest PSNR and SSIM values. Although the simulated cloud region is large and concentrated, the proposed method is still able to recover the missing information with considerable accuracy. In general, the simulated experiment shows that the proposed method achieved best results in both visual perception and objective quantitative indexes.

However, it should also be noted that the incorporation of both LRH image and SAR image has achieved the best performance. When only one of them is used as input, the cloud removal results are worse: the SAR image is very important to provide the detail structural information, whereas the LRH image is critical to recover land cover types correctly.

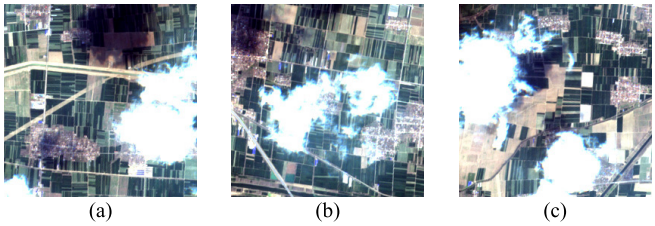


Fig. 11. Real cloudy HR images for experiment. (a) Cloud-contaminated HR image A. (b) Cloud-contaminated HR image B. (c) Cloud-contaminated HR image C.

B. Experimental Results on Real Cloud Removal

In this section, the proposed method is tested on a series of HR images with real cloud contaminations. The testing HR images for experiment are selected directly from the cloudy parts of region B in Fig. 4(a). The clouds are real thick clouds in the source image. Fig. 11 shows three cloud-contaminated HR images used for real cloud removal experiments.

Since the testing images are contaminated by real clouds, there are no ground truths. Therefore, the evaluation of the cloud removal performance is more subjective. On the one hand, the results should be evaluated by the similarity to the cloud-free regions. On the other hand, the corresponding LRH and SAR images still provide some useful information about the structural details under thick cloud cover.

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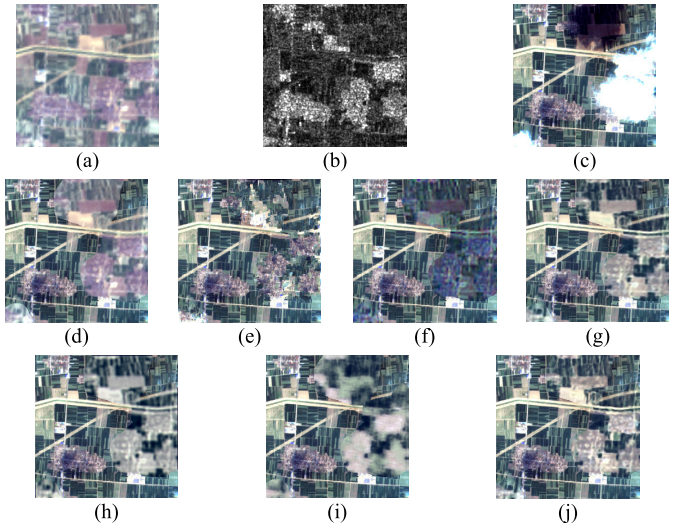


Fig. 12. Results of different methods in real cloud removal on image A. (a) LRH image. (b) SAR image. (c) Real cloudy HR image. (d) Result of DPL. (e) Result of SARSD. (f) Result of DGL. (g) Result of STS-CNN. (h) Result of CMD-LRH. (i) Result of CMD-SAR. (j) Result of CMD-LRH&SAR.

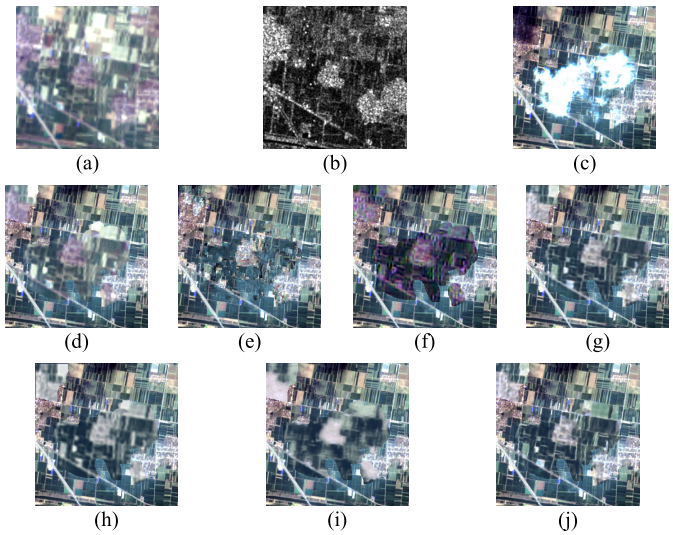


Fig. 13. Results of different methods in real cloud removal on image B. (a) LRH image. (b) SAR image. (c) Real cloudy HR image. (d) Result of DPL. (e) Result of SARSD. (f) Result of DGL. (g) Result of STS-CNN. (h) Result of CMD-LRH. (i) Result of CMD-SAR. (j) Result of CMD-LRH&SAR.

Figs. 12–14 show the corresponding cloud removal results of the real cloudy images in Fig. 11. It can be seen that the SARSD method failed to rebuild reasonable structural details in most cases. Although some missing information recovered by it seems to fit the background well, a lot of structural details are not recovered correctly. Both the DPL method and the DGL method show relatively lower similarity to the cloud-free regions in intensity, although they managed to restore more structural details. The STS-CNN method performed quite well in recovering the missing information, but some detail structural information is not perfect, especially continuous line structures. Our proposed method still provides the best performances in removing real cloud, recovering the missing information with both higher intensity similarity and structural details.

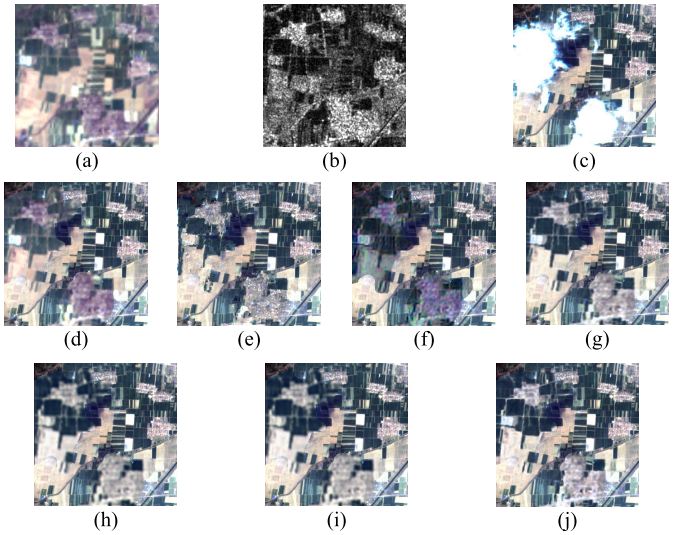


Fig. 14. Results of different methods in real cloud removal on image C. (a) LRH image. (b) SAR image. (c) Real cloudy HR image. (d) Result of DPL. (e) Result of SARSD. (f) Result of DGL. (g) Result of STS-CNN. (h) Result of CMD-LRH. (i) Result of CMD-SAR. (j) Result of CMD-LRH&SAR.

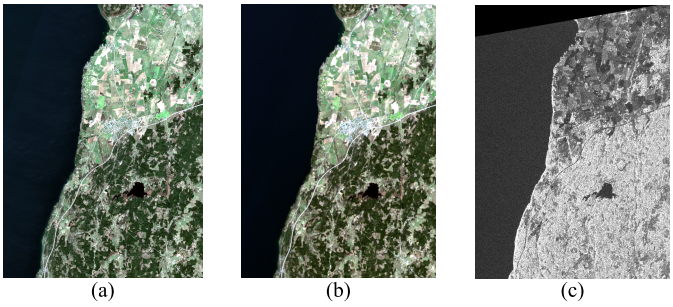


Fig. 15. Source images used for robustness testing. (a) HR source image (acquisition date: April 19, 2019). (b) LRH source image (acquisition date: April 17, 2019). (c) SAR source image (acquisition date: April 19, 2019).

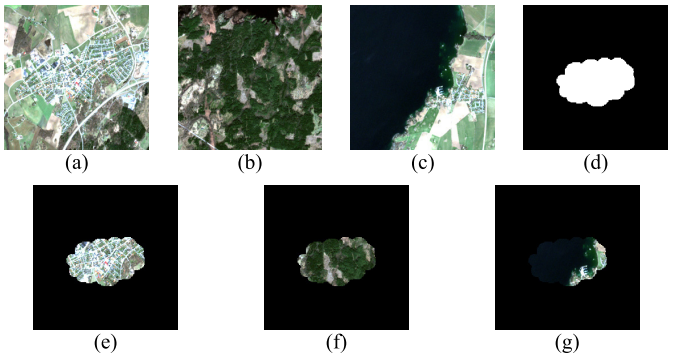


Fig. 16. Cloud-free HR images and cloud mask for simulated cloud removal. (a) Cloud-free HR image A. (b) Cloud-free HR image B. (c) Cloud-free HR image C. (d) Cloud mask. (e) Masked land covers in image A. (f) Masked land covers in image B. (g) Masked land covers in image C.

C. Experimental Results on Robustness

In order to test the robustness of our proposed method, we selected three testing samples from a different scene with land cover types not included in the training samples. The source images are shown in Fig. 15, and the spatial size is 1342×1842 pixels. This scene is located near the Vattern Lake in Sweden, in which the major land cover types are farmland, forest, and waterbody. Fig. 16 demonstrates the selected testing

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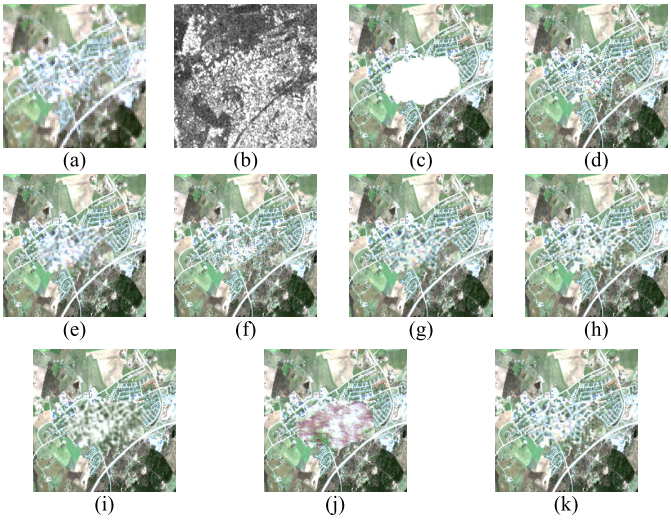


Fig. 17. Results of different methods in robustness testing on image A. (a) LRH image. (b) SAR image. (c) Simulated cloudy HR image. (d) Ground truth. (e) Result of DPL. (f) Result of SARSD. (g) Result of DGL. (h) Result of STS-CNN. (i) Result of CMD-LRH. (j) Result of CMD-SAR. (k) Result of CMD-LRH&SAR.

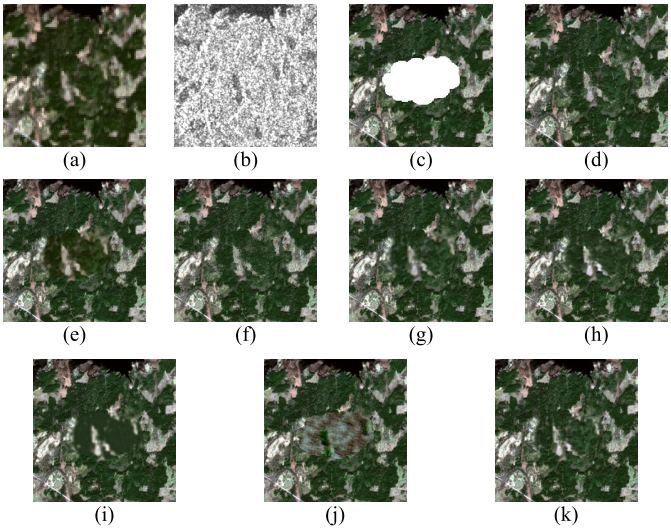


Fig. 18. Results of different methods in robustness testing on image B. (a) LRH image. (b) SAR image. (c) Simulated cloudy HR image. (d) Ground truth. (e) Result of DPL. (f) Result of SARSD. (g) Result of DGL. (h) Result of STS-CNN. (i) Result of CMD-LRH. (j) Result of CMD-SAR. (k) Result of CMD-LRH&SAR.

samples and the masked land covers in them. The masked land cover in image A is residential areas, in image B, it is forest, while in image C, it is lake waterbody. It should be noted that the land covers in training samples are mainly farmland and residential areas, including no forest and very little river waterbody with quite different appearance to the lake waterbody in image C. The cloud removal results are shown in Figs. 17–19 with the corresponding PSNR and SSIM values listed in Tables V–VII, respectively. It can be observed that CMD net still shows significant advantage in Fig. 17 by scoring the highest PSNR and SSIM values, since residential areas land cover is included in the training samples, although image A is from a totally different scene, the missing information can still be restored accurately. While in Fig. 18,

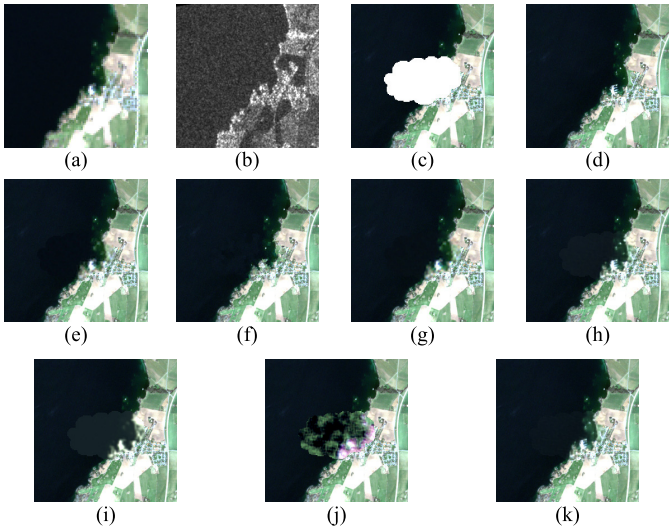


Fig. 19. Results of different methods in robustness testing on image C. (a) LRH image. (b) SAR image. (c) Simulated cloudy HR image. (d) Ground truth. (e) Result of DPL. (f) Result of SARSD. (g) Result of DGL. (h) Result of STS-CNN. (i) Result of CMD-LRH. (j) Result of CMD-SAR. (k) Result of CMD-LRH&SAR.

TABLE V

QUANTITATIVE EVALUATION FOR THE RESULTS IN FIG. 17

		DPL	SARSD	DGL	STS -CNN	CMD -LRH	CMD -SAR	CMD-LRH &SAR
PSNR	B4	25.23	22.82	26.24	26.37	25.36	20.02	**26.70**
	B3	25.74	23.32	26.34	26.47	25.14	20.69	**26.69**
	B2	24.04	21.95	24.84	25.72	24.07	19.43	**25.78**
SSIM	B4	0.9203	0.8883	0.9317	0.9476	0.9215	0.8562	**0.9509**
	B3	0.9166	0.8876	0.9246	0.9426	0.9139	0.8590	**0.9456**
	B2	0.9142	0.8870	0.9200	0.9466	0.9150	0.8559	**0.9473**

TABLE VI

QUANTITATIVE EVALUATION FOR THE RESULTS IN FIG. 18

		DPL	SARSD	DGL	STS -CNN	CMD -LRH	CMD -SAR	CMD-LRH &SAR
PSNR	B4	34.14	30.38	**34.74**	34.25	31.23	21.05	34.64
	B3	32.41	30.06	34.10	33.75	30.37	22.35	**34.17**
	B2	30.98	29.59	33.61	33.41	30.41	20.69	**33.65**
SSIM	B4	0.9602	0.9241	0.9628	**0.9757**	0.9449	0.8599	0.9733
	B3	0.9438	0.9091	0.9492	**0.9674**	0.9279	0.8597	0.9652
	B2	0.9380	0.9128	0.9503	**0.9718**	0.9291	0.8560	0.9689

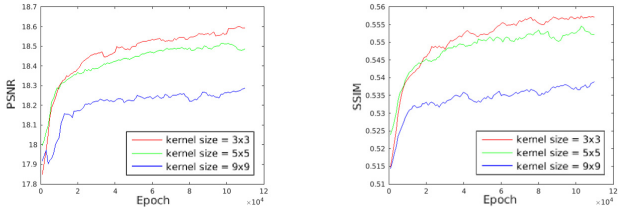
TABLE VII

QUANTITATIVE EVALUATION FOR THE RESULTS IN FIG. 19

		DPL	SARSD	DGL	STS -CNN	CMD -LRH	CMD -SAR	CMD-LRH &SAR
PSNR	B4	32.30	29.30	32.86	32.58	29.37	22.43	**33.12**
	B3	32.72	29.48	**32.79**	32.35	28.03	22.00	32.69
	B2	31.33	28.31	31.31	31.78	28.01	22.89	**31.86**
SSIM	B4	0.9741	0.9645	**0.9757**	0.9541	0.9233	0.8664	0.9668
	B3	0.9722	0.9553	0.9743	0.9743	0.9371	0.8602	**0.9795**
	B2	0.9570	0.9326	0.9562	0.9759	0.9369	0.8726	**0.9780**

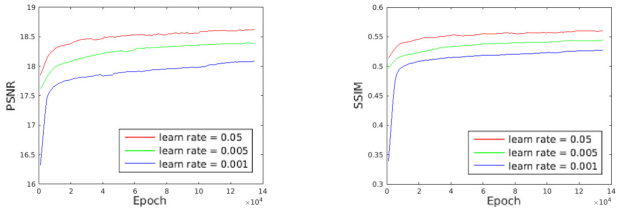
in which the masked land cover is forest and not included in the training samples, CMD net scores the highest PSNR values in bands 2 and 3. The highest SSIM values are scored by STS-CNN, and CMD net scores the second highest. This may be attributed to the successful multiscale design of STS-CNN, making it more robust to structural details. In Fig. 19, CMD net still scores most of the highest PSNR and SSIM

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(a) (b)

Fig. 20. PSNR and SSIM value growth influenced by kernel size. (a) PSNR versus epochs. (b) SSIM versus epochs.



(a) (b)

Fig. 21. PSNR and SSIM value growth influenced by learning rate. (a) PSNR versus epochs. (b) SSIM versus epochs.

values, even though the masked lake waterbody land cover is also not included in the training samples. In general, CMD net is able to handle data from other scenes as long as the land cover types are included in the training samples. When the type of land cover to be restored is not included in training data, the performance of CMD net is degraded. But compared to other approaches, CMD net still performs good enough under this extreme condition, which means that it has sufficient robustness.

D. Experimental Results on Parameter Sensitivity

In this section, we would demonstrate the influence of two most important parameters to the performance of the proposed method, namely the kernel size and the learning rate. The average PSNR and SSIM values of estimated images in 100 sets of cloud-free testing samples are used to quantize the ability of models trained with different parameters. It should be noted that the PSNR and SSIM values are calculated with the whole estimated results with no normalization, which means we assume all information in HR optical image is missing and intensity normalization is not available. We made this harsh assumption to suppress the interference of other factors such as the location of the cloud region and texture uniformity of Earth surface.

As we have mentioned, four layers with 3×3 kernels, two layers with 5×5 kernels, and one layer with 9×9 kernels share the same size of reception field. Their time cost is also close to each other. In general, decomposing one layer with 9×9 kernels into four layers with 3×3 kernels only increased the time cost by about 30%. However, the improvement in performance brought by it is significant. As shown in Fig. 20, models trained with 3×3 and 5×5 kernels show better performance and growth to the model trained with 9×9 kernels significantly. While the advantage of 3×3 kernels to 5×5 kernels is relatively limited, which means using 5×5 kernels can be a tactical option in practical application.

Different from the networks trained for natural image restoration, the value of learning rate influences not only the speed of convergence but also the final performance. As shown in Fig. 21, three models are trained with all the same parameters except for learning rate. In most deep learning models for natural image restoration, smaller learning rate usually means slower convergence but better final performance. However, in the proposed multisource estimation, smaller learning rate meant not only slowed down the convergence but also led to worse performance. The value of 0.05 is the largest available value we have tried. The whole network failed to converge when we increased the learning rate to 0.1.

In conclusion, both kernel size and learning rate are important parameters which would influence the performance of the CMD net. Experiments have demonstrated that smaller kernels outperform large kernels when the reception fields are the same. On the other hand, larger learning rate will influence the final performance of the CMD net. Larger learning rate would result in better performance, which is very different from networks for natural image restoration.

TABLE VIII
SPECTRAL BANDS FOR THE LANDSAT-8 SENSOR [37]

Spectral Band	Central wavelength (nm)	Bandwidth (nm)	Spatial resolution (m)
Band 1 - Coastal / Aerosol	443	20	30
Band 2 - Blue	483	65	30
Band 3 - Green	563	75	30
Band 4 - Red	655	50	30
Band 5 - Near Infrared	865	40	30
Band 6 - Short Wavelength Infrared	1610	100	30
Band 7 - Short Wavelength Infrared	2200	200	30
Band 8 - Panchromatic	590	180	15
Band 9 - Cirrus	1375	30	30
Band 10 - Long Wavelength Infrared	10800	1000	100
Band 11 - Long Wavelength Infrared	12000	1000	100

E. Experiment on More Matching Bands

In the proposed CMD net, only three visible bands are used, this is because the wavelengths of each band are expected to basically match in the LRH and HR images. As mentioned in [13], the introduced optical auxiliary source should have similar wavelength with the original one. Although the wavelengths of different bands vary among different optical sources, they are usually similar in three visible bands. Using three visible bands can ensure the basic matching of wavelengths between two different optical sources.

On the other hand, depending on the configurations of LRH and HR sources, there are usually other matching bands other than three visible bands. As shown in Tables VIII and IX,

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TABLE IX
SPECTRAL BANDS FOR THE SENTINEL-2A SENSOR [38]

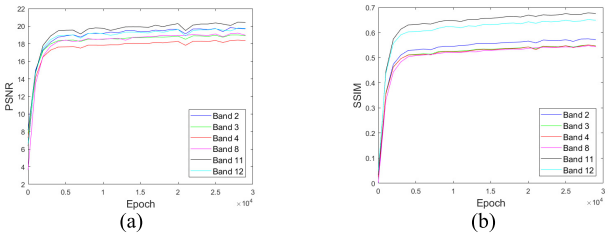
Spectral Band	Central wavelength (nm)	Bandwidth (nm)	Spatial resolution (m)
Band 1 – Coastal aerosol	442	21	60
Band 2 – Blue	492	66	10
Band 3 – Green	560	36	10
Band 4 – Red	665	31	10
Band 5 – Vegetation red edge	704	15	20
Band 6 – Vegetation red edge	741	15	20
Band 7 – Vegetation red edge	783	20	20
Band 8 – NIR	833	106	10
Band 8A – Narrow NIR	865	21	20
Band 9 – Water vapor	945	20	60
Band 10 – SWIR – Cirrus	1374	31	60
Band 11 – SWIR	1614	91	20

bands 5, 6, and 7 of Landsat-8 are considered to match bands 8, 11, and 12 of Sentinel-2, respectively. The spatial resolution of band 5 (Landsat-8) is 30 m, and for band 8 (Sentinel-2), it is 10 m, which is the same as the spatial resolution ratio in three visible bands. Although some difference exists between their central wavelengths, larger bandwidth of band 8 (Sentinel-2) enables it to cover the wavelength range of band 5 (Landsat-8). The wavelengths of bands 6 and 7 (Landsat-8) match those of bands 11 and 12 (Sentinel-2) well, but it should be noted that the spatial resolution in bands 11 and 12 (Sentinel-2) is 20 m, which means the spatial resolution ratio is different to bands 2, 3, 4, and 8 since the spatial resolution is still 30 m in bands 6 and 7 (Landsat-8).

For the rest of the bands of Sentinel-2, the spatial resolution of bands 1, 9, and 10 is 60 m, making them not suitable for HR source. Bands 5, 6, and 7 are vegetation red edge bands which have no correspondence in Landsat-8. Band 8A is a narrow NIR band, and its wavelength range is included by band 8.

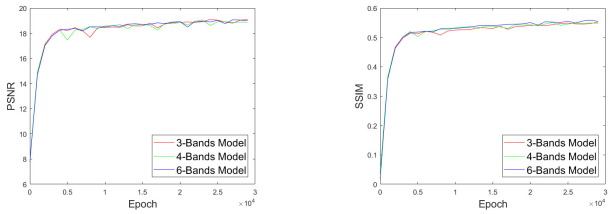
In this section, experiments are conducted to test the performance of the CMD net when processing more matching bands. Three models are trained with 1000 training samples from region C in Fig. 4(a): a six-band model taking bands 2, 3, 4, 5, 6, and 7 of Landsat-8 as input and producing bands 2, 3, 4, 8, 11, and 12 of Sentinel-2 as output, a four-band model taking bands 2, 3, 4, and 5 of Landsat-8 as input and producing bands 2, 3, 4, and 8 of Sentinel-2 as output, and a three-band model taking bands 2, 3, and 4 of Landsat-8 as input and producing bands 2, 3, and 4 of Sentinel-2 as output. Similar to Section IV-D, 100 sets of cloud-free testing samples are selected from region B in Fig. 4(a), and the average PSNR and SSIM values of the whole estimated results without normalization are used to evaluate the performance of three models.

Fig. 22 shows the PSNR and SSIM growth of different bands in the six-band model. It can be seen that the PSNR and SSIM values are higher than most visible bands in bands 11 and 12, but this is more likely to be attributed to their lower spatial resolution (20 m), since band 8 with the same spatial resolution (10 m) to visible bands also performs similar to visible bands.



(a) (b)

Fig. 22. PSNR and SSIM value growth of different bands in the six-band model. (a) PSNR versus epochs. (b) SSIM versus epochs.



(a) (b)

Fig. 23. Average PSNR and SSIM value growth of visible bands in different models. (a) PSNR versus epochs. (b) SSIM versus epochs.

When processing more matching bands with the CMD net, another important point is whether introducing more bands will degrade the performance of the original bands, especially when the spatial resolution of bands 11 and 12 is different from them. Fig. 23 shows the growth of average PSNR and SSIM values of three visible bands from three different models; it can be seen that the performances in visible bands are very similar among three-band model, four-band model, and six-band model, which means introducing more bands has little influence on the performance in the original visible bands.

Fig. 24 demonstrates one example of cloud removal results with six-band model. The first column shows the corresponding bands in the LRH image for the reference of land cover, the second column shows the cloud-contaminated HR image, and the third column shows the cloud removal result with six-band model. It can be seen that the missing information in each band can be recovered well, even if the radiation differs among different bands and the spatial resolutions are also not unified.

V. DISCUSSION

In the proposed CMD net, only three visible bands are used; this is because the wavelengths of each band are expected to basically match in LRH and HR image. As mentioned in [13], the introduced optical auxiliary source should have similar wavelength with the original one. Although the wavelengths of different bands vary among different optical sources, they are usually similar in three visible bands. Using three visible bands

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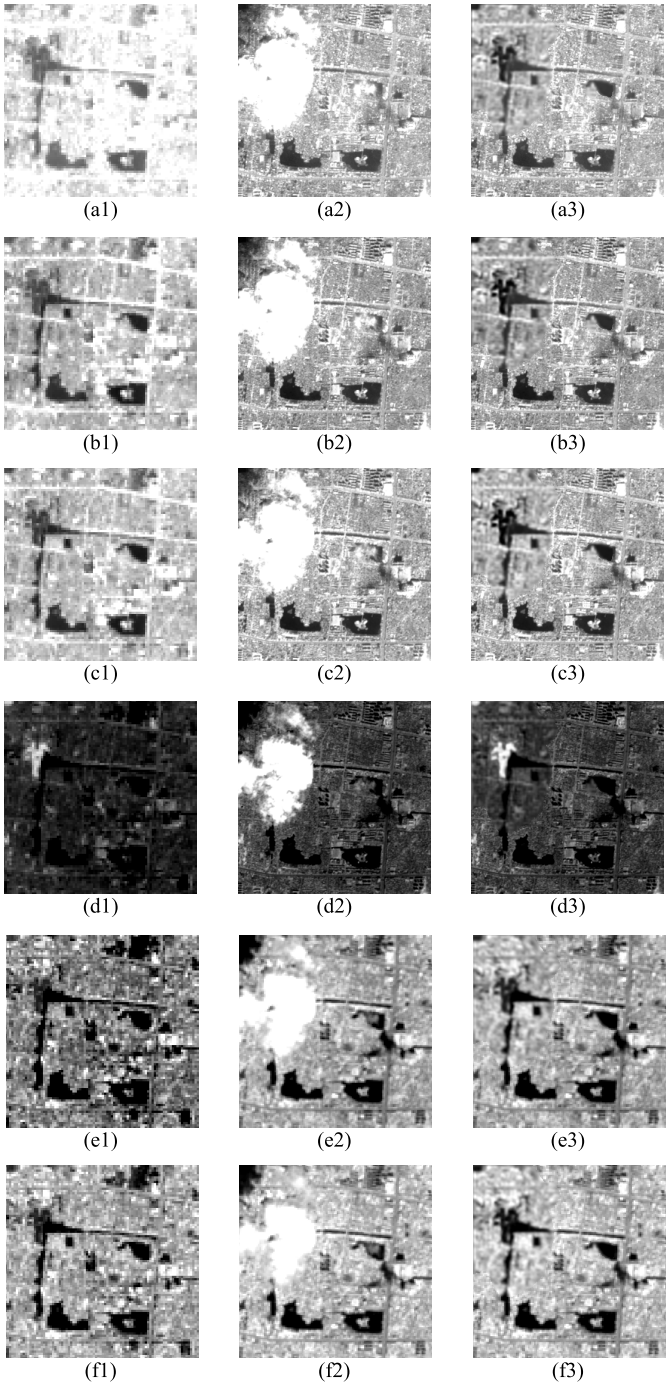


Fig. 24. Cloud removal results with six-band model. Column 1 shows bands 2, 3, 4, 5, 6, and 7 of LRH image as reference of land covers. Column 2 shows bands 2, 3, 4, 8, 11, and 12 of HR image contaminated by real clouds. Column 3 shows bands 2, 3, 4, 8, 11, and 12 of cloud removal result. (a1) Band 2 of LRH image. (a2) Band 2 of HR image. (a3) Band 2 of cloud removal result. (b1) Band 3 of LRH image. (b2) Band 3 of HR image. (b3) Band 3 of cloud removal result. (c1) Band 4 of LRH image. (c2) Band 4 of HR image. (c3) Band 4 of cloud removal result. (d1) Band 5 of LRH image. (d2) Band 8 of HR image. (d3) Band 8 of cloud removal result. (e1) Band 6 of LRH image. (e2) Band 11 of HR image. (e3) Band 11 of cloud removal result. (f1) Band 7 of LRH image. (f2) Band 12 of HR image. (f3) Band 12 of cloud removal result.

can ensure the basic matching of wavelengths between two different optical sources. On the other hand, our motivation to use Landsat-8, Sentinel-1, and Sentinel-2 data is that they all have free online access with global coverage. They only have three more matching bands apart from visible bands, i.e., bands 5, 6, and 7 of Landsat-8 and bands 8, 11, and 12 of Sentinel-2. We have demonstrated that CMD net is able to handle more bands without degrading the performance in original bands by adding these three extra matching bands to the model. When the CMD net is used to remove thick clouds from other products in which LRH source and HR source have more bands with similar wavelengths, it is possible to increase the number of optical bands to recover.

Being a deep learning-based approach, the performance of CMD net is also influenced by the training data. It will be degraded when restoring land cover types that are not included in the training samples. However, certain robustness still makes CMD net effective enough under this extreme condition. Since in most cases, the training samples are assumed to have covered most of the land cover types, the performance of CMD net is guaranteed. Another shortcoming of CMD net is that cloud region detection is required as a preprocessing step. But considering CMD net actually reconstructed the whole HR image instead of the cloud regions only, and the reconstruction is based on the information from LRH and SAR reference image, cloud-free regions are only used for intensity normalization, and the negative effect of inaccurate cloud region detection is also quite limited.

VI. Conclusion

In this article, we proposed a thick cloud removal method based on multisource estimation via the CMD net. The CMD net is composed of three functional components: the convolutional layers for encoding, the mapping layer for feature transferring, and the deconvolutional layers for decoding. LRH and SAR reference images are taken as input by the CMD net. The input images are first encoded into features by the convolutional layers, then transferred to the HR feature space by the mapping layer, and at last decoded into cloud-free HR output by the deconvolutional layers. Cloud-free pixels in the cloud-contaminated HR image are utilized to further improved the estimated cloud-free HR image via intensity normalization. At last, the cloudy pixels in the cloud-contaminated HR image are replaced by their corresponding pixels in the normalized cloud-free HR image. In this way, thick clouds are removed from the cloud-contaminated HR image. Experiments on removing both simulated and real clouds are carried out on data from real remote sensing sources. The results of our proposed method are compared with several recently proposed multisource-based cloud removal methods. Significant advantages of our proposed method are demonstrated by the comparison. Samples selected from a different scene with land cover types not included in training samples are used to validate the robustness of the CMD net. Experiments for important parameters are also made, and the results show that both smaller kernel size and larger learning rate are beneficial to improving the result of estimation. The experiments on more matching bands between Landsat-8 and Sentinel-2 demonstrate that CMD net is able to handle more bands without degrading the performance of original bands.

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On the other hand, some new challenges are also posed by this work. First, the training of CMD net for multisource estimation is not as steady as image restoration. A large learning rate is required to ensure the speed of convergence and performance of the network. Future works should be done to build a steadier mapping between multisource remote sensing images. Second, the usage of LRH imagery seems to be redundant since it is acquired with optical sensors similar to the HR imagery. It is also a challenge to estimate the HR data directly from SAR data.

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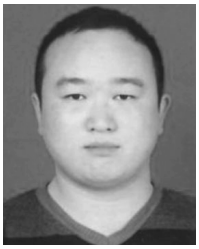
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